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Windows Forensic Analysis

POSTER

You Can't Protect What You Don't Know About

digital-forensics.sans.org

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Poster Created by Rob Lee with support of the SANS DFIR Faculty
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Windows® Time Rules								
STANDARD_INFORMATION								
File Creation	File Access	File Modification	File Rename	File Copy	Local File Move	Volume File Move (move via CLI)	Volume File Move (cut/paste via Explorer)	File Deletion
Modified – Time of File Creation	Modified – No Change	Modified – Time of Data Modification	Modified – No Change	Modified – Inherited from Original	Modified – No Change	Modified – Inherited from Original	Modified – Inherited from Original	Modified – No Change
Access – Time of File Creation	Access – Time of Access (No Change only on NTFS Win7+)	Access – No Change	Access – No Change	Access – Time of File Copy	Access – No Change	Access – Time of File Move via CLI	Access – Time of Cut/Paste	Access – No Change
Metadata – Time of File Creation	Metadata – No Change	Metadata – Time of Data Modification	Metadata – Time of File Rename	Metadata – Time of File Copy	Metadata – Time of Local File Move	Metadata – Inherited from Original	Metadata – Inherited from Original	Metadata – No Change
Creation – Time of File Creation	Creation – No Change	Creation – No Change	Creation – No Change	Creation – Time of File Copy	Creation – No Change	Creation – Time of File Move via CLI	Creation – Inherited from Original	Creation – No Change

FILE NAME								
File Creation	File Access	File Modification	File Rename	File Copy	Local File Move	Volume File Move (move via CLI)	Volume File Move (cut/paste via Explorer)	File Deletion
Modified – Time of File Creation	Modified – No Change	Modified – No Change	Modified – No Change	Modified – Time of File Copy	Modified – No Change	Modified – Time of Move via CLI	Modified – Time of Cut/Paste	Modified – No Change
Access – Time of File Creation	Access – No Change	Access – No Change	Access – No Change	Access – Time of File Copy	Access – No Change	Access – Time of Move via CLI	Access – Time of Cut/Paste	Access – No Change
Metadata – Time of File Creation	Metadata – No Change	Metadata – No Change	Metadata – No Change	Metadata – Time of File Copy	Metadata – No Change	Metadata – Time of Move via CLI	Metadata – Time of Cut/Paste	Metadata – No Change
Creation – Time of File Creation	Creation – No Change	Creation – No Change	Creation – No Change	Creation – Time of File Copy	Creation – No Change	Creation – Time of Move via CLI	Creation – Time of Cut/Paste	Creation – No Change



Windows Artifact Analysis: Evidence of...

The “Evidence of...” categories were originally created by SANS Digital Forensics and Incidence Response faculty for the SANS course FOR500: Windows Forensic Analysis. The categories map a specific artifact to the analysis questions that it will help to answer. Use this poster as a cheat-sheet to help you remember where you can discover key Windows artifacts for computer intrusion, intellectual property theft, and other common cyber crime investigations.

File Download

Open/Save MRU

Description

In the simplest terms, this key tracks files that have been opened or saved within a Windows shell dialog box. This happens to be a big data set, not only including web browsers like Internet Explorer and Firefox, but also a majority of commonly used applications.

Location

XP:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU
Win7/8/10:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePIDMRU

Interpretation

- The “*” key – This subkey tracks the most recent files of any extension input in an OpenSave dialog
- “???” (Three letter extension) – This subkey stores file info from the OpenSave dialog by specific extension

Email Attachments

Description

The email industry estimates that 80% of email data is stored via attachments. Email standards only allow text. Attachments must be encoded with MIME/base64 format.

Location

Outlook
XP:
%SERPROFILE%\Local Settings\ApplicationData\Microsoft\Outlook
Win7/8/10:
%USERPROFILE%\AppData\Local\Microsoft\Outlook

Interpretation

MS Outlook data files found in these locations include OST and PST files. One should also check the OLK and Content.Outlook folder, which might roam depending on the specific version of Outlook used. For more information on where to find the OLK folder this link has a handy chart: <http://www.hanockcomputerstech.com/blog/2010/01/06/find-the-microsoft-outlook-temporary-olk-folder>

Skype History

Description

Skype history keeps a log of chat sessions and files transferred from one machine to another
- This is turned on by default in Skype installations

Location

XP:
C:\Documents and Settings<username>\Application\Skype\<skype-name>
Win7/8/10:
C:\%USERPROFILE%\AppData\Roaming\Skype\<skype-name>

Interpretation

Each entry will have a date/time value and a Skype username associated with the action.

Browser Artifacts

Description

Not directly related to “File Download”. Details stored for each local user account. Records number of times visited (frequency).

Location

Internet Explorer
• IE8-9:
%USERPROFILE%\AppData\Roaming\Microsoft\Windows\IEDownloadHistory\index.dat
• IE10-11:
%USERPROFILE%\AppData\Local\Microsoft\Windows\WebCache\WebCacheV9.dat
Firefox
• V3-25:
%userprofile%\AppData\Roaming\Mozilla\Firefox\Profiles<random text>\.default\downloads.sqlite
• v26+:
%userprofile%\AppData\Roaming\Mozilla\Firefox\Profiles<random text>\.default\places.sqlite
Table.moz annos
Chrome:
• Win7/8/10:
%USERPROFILE%\AppData\Local\Google\Chrome\User Data\Default\History

Interpretation

Many sites in history will list the files that were opened from remote sites and downloaded to the local system. History will record the access to the file on the website that was accessed via a link.

Downloads

Description

Firefox and IE have a built-in download manager application which keeps a history of every file downloaded by the user. This browser artifact can provide excellent information about what sites a user has been visiting and what kinds of files they have been downloading from them.

Location

Firefox:
• XP:
%userprofile%\Application Data\Mozilla\Firefox\Profiles<random text>\.default\downloads.sqlite
• Win7/8/10:
%userprofile%\AppData\Roaming\Mozilla\Firefox\Profiles<random text>\.default\downloads.sqlite
Internet Explorer:
• IE8-9:
%USERPROFILE%\AppData\Roaming\Microsoft\Windows\IEDownloadHistory
• IE10-11:
%USERPROFILE%\AppData\Local\Microsoft\Windows\WebCache\WebCacheV9.dat

Interpretation

Downloads will include:
• Filename, Size, and Type
• Download from and Referring Page
• File Save Location
• Application Used to Open File
• Download Start and End Times

ADS Zone.Identifier

Description

Starting with XP SP2 when files are downloaded from the “Internet Zone” via a browser to a NTFS volume, an alternate data stream is added to the file. The alternate data stream is named “Zone.Identifier.”

Interpretation

Files with an ADS Zone.Identifier and contains ZoneID=3 were downloaded from the Internet
• URLZONE_TRUSTED = ZoneID = 2
• URLZONE_INTERNET = ZoneID = 3
• URLZONE_UNTRUSTED = ZoneID = 4

Program Execution

UserAssist

Description

GUI-based programs launched from the desktop are tracked in the launcher on a Windows System.

Location

NTUSER.DAT HIVE:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{GUID}\Count

Interpretation

All values are ROT-13 Encoded
• GUID for XP
- 75048700 Active Desktop
• GUID for Win7/8/10
- CEBFF5CD Executable File Execution
- F4E57C4B Shortcut File Execution

Windows 10 Timeline

Description

Win10 records recently used applications and files in a “timeline” accessible via the “WIN+TAB” key. The data is recorded in a SQLite database.

Location

C:\Users\<profile>\AppData\Local\ConnectedDevicesPlatform\L\<profile>\ActivitiesCache.db

Interpretation

• Application execution
• Focus count per application

RecentApps

Description

GUI Program execution launched on the Win10 system is tracked in the RecentApps key

Location

Win10:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Search\RecentApps

Interpretation

Each GUID key points to a recent application.
AppID = Name of Application
LastAccessTime = Last execution time in UTC
LaunchCount = Number of times executed

Shimcache

Description

Windows Application Compatibility Database is used by Windows to identify possible application compatibility challenges with executables.
• Tracks the executables file name, file size, last modified time, and in Windows XP the last update time

Location

XP:
SYSTEM\CurrentControlSet\Control\SessionManager\AppCompatibility
Win7/8/10:
SYSTEM\CurrentControlSet\Control\SessionManager\AppCompatCache

Interpretation

Any executable run on the Windows system could be found in this key. You can use this key to identify systems that specific malware was executed on. In addition, based on the interpretation of the time-based data you might be able to determine the last time of execution or activity on the system.
• Windows XP contains at most 96 entries
- LastUpdateTime is updated when the files are executed
• Windows 7 contains at most 1,024 entries
- LastUpdateTime does not exist on Win7 systems

Jump Lists

Description

The Windows 7 task bar (Jump List) is engineered to allow users to “jump” or access items they have frequently or recently used quickly and easily. This functionality cannot only include recent media files; it must also include recent tasks.
• The data stored in the AutomaticDestinations folder will each have a unique file prepended with the AppID of the associated application.

Location

Win7/8/10:
C:\%USERPROFILE%\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations

Interpretation

• First time of execution of application.
- Creation Time = First time item added to the AppID file.
• Last time of execution of application w/ file open.
- Modification Time = Last time item added to the AppID file.
• List of Jump List IDs ->
http://www.forensicswiki.org/wiki/List_of_Jump_List_IDS

Amcache.hve

Description

ProgramDataUpdater (a task associated with the Application Experience Service) uses the registry file Amcache.hve to store data during process creation

Location

Win7/8/10:
C:\Windows\AppCompat\Programs\Amcache.hve

Interpretation

• Amcache.hve – Keys = Amcache.hve\Root\File\Volume GUID\#####
• Entry for every executable run, full path information, File’s \$StandardInfo Last Modification Time, and Disk volume the executable was run from
• First Run Time = Last Modification Time of Key
• SHA1 hash of executable also contained in the key

System Resource Usage Monitor (SRUM)

Description

Records 30 to 60 days of historical system performance. Applications run, user account responsible for each, and application and bytes sent/received per application per hour.

Location

SOFTWARE\Microsoft\Windows\NT\CurrentVersion\SRUM\Extensions\{d10ca26e-6fcf-4f6d-848e-b2e99266fa89} = Application Resource Usage Provider C:\Windows\System32\SRU

Interpretation

Use tool such as srum_dump.exe to cross correlate the data between the registry keys and the SRUM ESE Database.

BAM/DAM

Description

Windows Background Activity Moderator (BAM)

Location

Win10:
SYSTEM\CurrentControlSet\Services\bam\UserSettings\SID
SYSTEM\CurrentControlSet\Services\dsm\UserSettings\SID

Investigative Notes

Provides full path of the executable file that was run on the system and last execution date/time

Last-Visited MRU

Description

Tracks the specific executable used by an application to open the files documented in the OpenSaveMRU key. In addition, each value also tracks the directory location for the last file that was accessed by that application.

Location

XP:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedMRU
Win7/8/10:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidlMRU

Interpretation

Tracks the application executables used to open files in OpenSaveMRU and the last file path used.

Prefetch

Description

Increases performance of a system by pre-loading code pages of commonly used applications. Cache Manager monitors all files and directories referenced for each application or process and maps them into a .pf file. Utilized to know an application was executed on a system.
• Limited to 128 files on XP and Win7
• Limited to 1024 files on Win8
• (exename)-(hash).pf

Location

WinXP/7/8/10:
C:\Windows\Prefetch

Interpretation

• Each .pf will include last time of execution, number of times run, and device and file handles used by the program
• Date/Time file by that name and path was first executed - Creation Date of .pf file (-10 seconds)
• Date/Time file by that name and path was last executed - Embedded last execution time of .pf file
• Last modification date of .pf file (-10 seconds)
• Win8-10 will contain last 8 times of execution

Deleted File or File Knowledge

XP Search – ACMRU

Description

You can search for a wide range of information through the search assistant on a Windows XP machine. The search assistant will remember a user’s search terms for filenames, computers, or words that are inside a file. This is an example of where you can find the “Search History” on the Windows system.

Location

WinXP/Win8/8.1
NTUSER.DAT HIVE
NTUSER.DAT\Software\Microsoft\Search Assistant\ACMRU\###

Interpretation

• Search the Internet – #####5001
• All or part of a document name – #####5603
• A word or phrase in a file – #####5604
• Printers, Computers and People – #####5647

Thumbs.db

Description

Hidden file in directory where images on machine exist stored in a smaller thumbnail graphics. thumbs.db catalogs pictures in a folder and stores a copy of the thumbnail even if the pictures were deleted.

Location

WinXP/Win8/8.1
Automatically created anywhere with homegroup enabled
Win7/8/10
Automatically created anywhere and accessed via a UNC Path (local or remote)

Interpretation

Include:
• Thumbnail Picture of Original Picture
• Document Thumbnail – Even if Deleted
• Last Modification Time (XP Only)
• Original Filename (XP Only)

Search – WordWheelQuery

Description

Keywords searched for from the START menu bar on a Windows 7 machine.

Location

Win7/8/10 NTUSER.DAT HIVE
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery

Interpretation

Keywords are added in Unicode and listed in temporal order in an MRUlist

Win7/8/10 Recycle Bin

Description

The recycle bin is a very important location on a Windows file system to understand. It can help you when accomplishing a forensic investigation, as every file that is deleted from a Windows recycle bin aware program is generally first put in the recycle bin.

Location

Hidden System Folder
Win7/8/10
• C:\\$Recycle.bin
• Deleted Time and Original Filename contained in separate files for each deleted recovery file

Interpretation

• SID can be mapped to user via Registry Analysis
• Win7/8/10
- Files Preceded by \$I##### files contain
• Original PATH and name
• Deletion Date/Time
- Files Preceded by \$R##### files contain
• Recovery Data

Last-Visited MRU

Description

Tracks the specific executable used by an application to open the files documented in the OpenSaveMRU key. In addition, each value also tracks the directory location for the last file that was accessed by that application.

Location

XP
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedMRU
Win7/8/10
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidlMRU

Interpretation

Tracks the application executables used to open files in OpenSaveMRU and the last file path used.

XP Recycle Bin

Description

The recycle bin is a very important location on a Windows file system to understand. It can help you when accomplishing a forensic investigation, as every file that is deleted from a Windows recycle bin aware program is generally first put in the recycle bin.

Location

Hidden System Folder
Windows XP
• C:\RECYCLER\2000\NT\XP\2003
• Subfolder is created with user’s SID
• Hidden file in directory called “INFO2”
• INFO2 Contains Deleted Time and Original Filename
• Filename in both ASCII and UNICODE

Interpretation

• SID can be mapped to user via Registry Analysis
• Maps file name to the actual name and path it was deleted from

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dfir.to/MAIL-LIST



dfir.to/DFIRCast

OPERATING SYSTEM & DEVICE IN-DEPTH



FOR498
Battlefield
Forensics & Data
Acquisition



FOR526
Advanced Memory
Forensics &
Threat Detection



FOR500
Windows Forensics



FOR585
Smartphone
Forensic Analysis
In-Depth



FOR518
Mac and iOS
Forensic Analysis
and Incident
Response

INCIDENT RESPONSE & THREAT HUNTING



FOR508
Advanced Incident
Response, Threat
Hunting, and Digital
Forensics



FOR572
Advanced Network
Forensics: Threat Hunting,
Analysis, and Incident
Response



FOR578
Cyber Threat Intelligence



FOR610
REM: Malware Analysis



SEC504
Hacker Tools, Techniques,
Exploits, and Incident Handling



Network Activity/Physical Location

Timezone	Network History	Browser Search Terms
Description Identifies the current system time zone. Location SYSTEM Hive: SYSTEM\CurrentControlSet\Control\TimeZoneInformation Interpretation - Time activity is incredibly useful for correlation of activity - Internal log files and date/timestamps will be based on the system time zone information - You might have other network devices and you will need to correlate information to the time zone information collected here.	Description - Identify networks that the computer has been connected to - Networks could be wireless or wired - Identify domain name/intranet name - Identify SSID - Identify Gateway MAC Address Location Win7/8/10: SOFTWARE HIVE: - SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures\Unmanaged - SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures\Managed - SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Nla\Cache Interpretation - Identifying intranets and networks that a computer has connected to is incredibly important - Not only can you determine the intranet name, you can determine the last time the network was connected to it based on the last write time of the key - This will also list any networks that have been connected to via a VPN - MAC Address of SSID for Gateway could be physically triangulated	Description Records websites visited by date and time. Details stored for each local user account. Records number of times visited (frequency). Also tracks access of local system files. This will also include the website history of search terms in search engines. Location Internet Explorer - IE6-7: %USERPROFILE%\Local Settings\History\History.IE5 - IE8-9: %USERPROFILE%\AppData\Local\Microsoft\Windows\History\History.IE5 - IE10-11: %USERPROFILE%\AppData\Local\Microsoft\Windows\WebCache\WebCacheV9.dat Firefox - XP: %userprofile%\Application Data\Mozilla\Firefox\Profiles\<randomtext>.default\places.sqlite - Win7/8/10: %userprofile%\AppData\Roaming\Mozilla\Firefox\Profiles\<randomtext>.default\places.sqlite System Resource Usage Monitor (SRUM) Description Records 30 to 60 days of historical system performance. Applications run, user account responsible for each, and application and bytes sent/received per application per hour. Location SOFTWARE\Microsoft\Windows NT\CurrentVersion\SRUM\Extensions (973f5d5c-1090-4944-8b8e-24b94231a74) = Windows Network Data Usage Monitor (db6636c4-8929-4683-974e-22c0464a43763) = Windows Network Connectivity Usage Monitor SOFTWARE\Microsoft\WlanSvc\Interfaces\ C:\Windows\System32\SRU\ Interpretation Use tool such as srum_dump.exe to cross correlate the data between the registry keys and the SRUM ESE Database.
Cookies	WLAN Event Log	
Description Cookies give insight into what websites have been visited and what activities may have taken place there. Location Internet Explorer - IE6-8: %USERPROFILE%\AppData\Roaming\Microsoft\Windows\Cookies - IE10: %USERPROFILE%\AppData\Roaming\Microsoft\Windows\Cookies - IE11: %USERPROFILE%\AppData\Local\Microsoft\Windows\NetCookies Firefox - XP: %USERPROFILE%\Application Data\Mozilla\Firefox\Profiles\<random text>.default\cookies.sqlite - Win7/8/10: %USERPROFILE%\AppData\Roaming\Mozilla\Firefox\Profiles\<randomtext>.default\cookies.sqlite Chrome - XP: %USERPROFILE%\Local Settings\Application Data\Google\Chrome\User Data\Default\Local Storage - Win7/8/10: %USERPROFILE%\AppData\Local\Google\Chrome\User Data\Default\Local Storage	Description Determine what wireless networks the system associated with and identify network characteristics to find location Relevant Event IDs 11000 – Wireless network association started 8001 – Successful connection to wireless network 8002 – Failed connection to wireless network 8003 – Disconnect from wireless network 6100 – Network diagnostics (System log) Location Microsoft-Windows-WLAN-AutoConfig Operational.evtx Interpretation - Shows historical record of wireless network connections - Contains SSID and BSSID (MAC address), which can be used to geolocate wireless access point *(no BSSID on Win8+)	



File/Folder Opening

Open/Save MRU	Shell Bags	Last-Visited MRU
Description In the simplest terms, this key tracks files that have been opened or saved within a Windows shell dialog box. This happens to be a big data set, not only including web browsers like Internet Explorer and Firefox, but also a majority of commonly used applications. Location XP: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU Win7/8/10: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePIDIMRU Interpretation - The "*" key – This subkey tracks the most recent files of any extension input in an OpenSave dialog *** (Three letter extension) – This subkey stores file info from the OpenSave dialog by specific extension	Description - Which folders were accessed on the local machine, the network, and/or removable devices. Evidence of previously existing folders after deletion/overwrite. When certain folders were accessed. Location Explorer Access: - USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\Bags - USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\BagMRU Desktop Access: - NTUSER.DAT\Software\Microsoft\Windows\Shell\BagMRU - NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags Interpretation Stores information about which folders were most recently browsed by the user. Shortcut (LNK) Files Description - Shortcut Files automatically created by Windows - Recent Items - Opening local and remote data files and documents will generate a shortcut file (.lnk) Location XP: C:\%USERPROFILE%\Recent Win7/8/10: C:\%USERPROFILE%\AppData\Roaming\Microsoft\Windows\Recent\ NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs Interpretation - RecentDocs – Overall key will track the overall order of the last 150 files or folders opened. MRU list will keep track of the temporal order in which each file/folder was opened. The last entry and modification time of this key will be the time and location the last file of a specific extension was opened. *** – This subkey stores the last files with a specific extension that were opened. MRU list will keep track of the temporal order in which each folder was opened. The last entry and modification time of this key will be the time when and location where the last file of a specific extension was opened. - Foilder – This subkey stores the last folders that were opened. MRU list will keep track of the temporal order in which each folder was opened. The last entry and modification time of this key will be the time and location of the last folder opened.	Description Tracks the specific executable used by an application to open the files documented in the OpenSaveMRU key. In addition, each value also tracks the directory location for the last file that was accessed by that application. Example: Notepad.exe was last run using the C:\Users\Rob\Desktop folder Location XP: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedMRU Win7/8/10: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidimRU Interpretation Tracks the application executables used to open files in OpenSaveMRU and the last file path used. IE Edge file:// Description A little known fact about the IE History is that the information stored in the history files is not just related to Internet browsing. The history also records local, removable, and remote (via network shares) file access, giving us an excellent means for determining which files and applications were accessed on the system, day by day. Location Internet Explorer: - IE6-7: %USERPROFILE%\Local Settings\History\History.IE5 - IE8-9: %USERPROFILE%\AppData\Local\Microsoft\Windows\History\History.IE5 - IE10-11: %USERPROFILE%\AppData\Local\Microsoft\Windows\WebCache\WebCacheV9.dat Interpretation - Stored in index.dat as: file://C:/directory/filename.ext - Does not mean file was opened in browser
Jump Lists	Prefetch	Office Recent Files
Description - The Windows 7 task bar (Jump List) is engineered to allow users to "jump" or access items have frequently or recently used quickly and easily. This functionality cannot only include recent media files; it must also include recent tasks. - The data stored in the AutomaticDestinations folder will each have a unique file prepended with the AppID of the association application and embedded with LNK files in each stream. Location Win7/8/10: C:\%USERPROFILE%\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations Interpretation - Using the Structured Storage Viewer, open up one of the AutomaticDestination jumplist files. - Each one of these files is a separate LNK file. They are also stored numerically in order from the earliest one (usually 1) to the most recent (largest integer value).	Description - Increases performance of a system by pre-loading code pages of commonly used applications. Cache Manager monitors all files and directories referenced for each application or process and maps them into a .pf file. Utilized to know an application was executed on a system. - Limited to 128 files on XP and Win7 - Limited to 1024 files on Win8-10 (exename)-(hash).pf Location WinXP/7/8/10: C:\Windows\Prefetch Interpretation - Can examine each .pf file to look for file handles recently used - Can examine each .pf file to look for device handles recently used	Description MS Office programs will track their own Recent Files list to make it easier for users to remember the last file they were editing. Location NTUSER.DAT\Software\Microsoft\Office\VERSION 14.0 = Office 2010 11.0 = Office 2003 12.0 = Office 2007 10.0 = Office XP NTUSER.DAT\Software\Microsoft\Office\VERSION\UserMRU\LiveID_###\FileMRU 15.0 = Office 365 Interpretation Similar to the Recent Files, this will track the last files that were opened by each MS Office application. The last entry added, per the MRU, will be the time the last file was opened by a specific MS Office application.

Last Login	Logon Types																								
Description Lists the local accounts of the system and their equivalent security identifiers. Location - C:\windows\system32\config\SAM - SAM\Domains\Account\Users Interpretation - Only the last login time will be stored in the registry key	Description Logon Events can give us very specific information regarding the nature of account authorizations on a system if we know where to look and how to decipher the data that we find. In addition to telling us the date, time, username, hostname, and success/failure status of a logon, Logon Events also enables us to determine by exactly what means a logon was attempted. Location Win7/8/10: Event ID 4624 Interpretation <table><thead><tr><th>Logon Type</th><th>Explanation</th></tr></thead><tbody><tr><td>2</td><td>Logon via console</td></tr><tr><td>3</td><td>Network Logon</td></tr><tr><td>4</td><td>Batch Logon</td></tr><tr><td>5</td><td>Windows Service Logon</td></tr><tr><td>7</td><td>Credentials used to unlock screen</td></tr><tr><td>8</td><td>Network logon sending credentials (cleartext)</td></tr><tr><td>9</td><td>Different credentials used than logged on user</td></tr><tr><td>10</td><td>Remote interactive logon (RDP)</td></tr><tr><td>11</td><td>Cached credentials used to logon</td></tr><tr><td>12</td><td>Cached remote interactive (similar to Type 10)</td></tr><tr><td>13</td><td>Cached unlock (similar to Type 7)</td></tr></tbody></table>	Logon Type	Explanation	2	Logon via console	3	Network Logon	4	Batch Logon	5	Windows Service Logon	7	Credentials used to unlock screen	8	Network logon sending credentials (cleartext)	9	Different credentials used than logged on user	10	Remote interactive logon (RDP)	11	Cached credentials used to logon	12	Cached remote interactive (similar to Type 10)	13	Cached unlock (similar to Type 7)
Logon Type	Explanation																								
2	Logon via console																								
3	Network Logon																								
4	Batch Logon																								
5	Windows Service Logon																								
7	Credentials used to unlock screen																								
8	Network logon sending credentials (cleartext)																								
9	Different credentials used than logged on user																								
10	Remote interactive logon (RDP)																								
11	Cached credentials used to logon																								
12	Cached remote interactive (similar to Type 10)																								
13	Cached unlock (similar to Type 7)																								
RDP Usage	Authentication Events																								
Description Track Remote Desktop Protocol logons to target machines. Location Security Log Win7/8/10: %SYSTEM ROOT%\System32\winevt\logs\Security.evtx Interpretation - Win7/8/10 – Interpretation - Event ID 4778 – Session Connected/Reconnected - Event ID 4779 – Session Disconnected - Event log provides hostname and IP address of remote machine making the connection - On workstations you will often see current console session disconnected (4779) followed by RDP connection (4778)	Description Authentication mechanisms Location Recorded on system that authenticated credentials Local Account/Workgroup = on workstation Domain/Active Directory = on domain controller Win7/8/10: %SYSTEM ROOT%\System32\winevt\logs\Security.evtx Interpretation Event ID Codes (NTLM protocol) 4776: Successful/Failed account authentication Event ID Codes (Kerberos protocol) 4768: Ticket Granting Ticket was granted (successful logon) 4769: Service Ticket requested (access to server resource) 4771: Pre-authentication failed (failed logon)																								
Services Events	Success/Fail Logons																								
Description - Analyze logs for suspicious services running at boot time - Review services started or stopped around the time of a suspected compromise Location All Event IDs reference the System Log 7034 – Service crashed unexpectedly 7035 – Service sent a Start/Stop control 7036 – Service started or stopped 7040 – Start type changed (Boot On Request Disabled) 7045 – A service was installed on the system (Win2008R2+) 4697 – A service was installed on the system (from Security log) Interpretation - All Event IDs except 4697 reference the System Log - A large amount of malware and worms in the wild utilize Services - Services started on boot illustrate persistence (desirable in malware) - Services can crash due to attacks like process injection	Description Determine which accounts have been used for attempted logons. Track account usage for known compromised accounts. Location Win7/8/10: %system root%\System32\winevt\logs\Security.evtx Interpretation - Win7/8/10 – Interpretation 4624 – Successful Logon 4625 – Failed Logon 4634 4647 – Successful Logoff 4648 – Logon using explicit credentials (Runas) 4672 – Account logon with superuser rights (Administrator) 4720 – An account was created																								

External Device/USB Usage

Key Identification	PnP Events	Drive Letter and Volume Name
Description Track USB devices plugged into a machine. Location - SYSTEM\CurrentControlSet\Enum\USBSTOR - SYSTEM\CurrentControlSet\Enum\USB Interpretation - Identify vendor, product, and version of a USB device plugged into a machine - Identify a unique USB device plugged into the machine - Determine the time a device was plugged into the machine - Devices that do not have a unique serial number will have an "&" in the second character of the serial number.	Description When a Plug and Play driver install is attempted, the service will log an ID 20001 event and provide a Status within the event. It is important to note that this event will trigger for any Plug and Play-capable device, including but not limited to USB, Firewire, and PCMCIA devices. Location System Log File Win7/8/10: %system root%\System32\winevt\logs\System.evtx Interpretation - Event ID: 20001 – Plug and Play driver install attempted - Event ID 20001 - Timestamp - Device information - Device serial number - Status (0 = no errors)	Description Discover the last drive letter of the USB Device when it was plugged into the machine. Location XP: Find ParentIdPrefix – SYSTEM\CurrentControlSet\Enum\USBSTOR Using ParentIdPrefix: Discover Last Mount Point – SYSTEM\MountedDevices Win7/8/10: SOFTWARE\Microsoft\Windows Portable Devices\Devices - SYSTEM\MountedDevices - Examine Drive Letters looking at Value Data Looking for Serial Number Interpretation Identify the USB device that was last mapped to a specific drive letter. This technique will only work for the last drive mapped. It does not contain historical records of every drive letter mapped to a removable drive.
First/Last Times	Volume Serial Number	Shortcut (LNK) Files
Description Determine temporal usage of specific USB devices connected to a Windows Machine. Location First Time Plug and Play Log Files XP: C:\Windows\setupapi.log Win7/8/10: C:\Windows\inf\setupapi.dev.log Interpretation - Search for Device Serial Number - Log File times are set to local time zone Location First, Last, and Removal Times (Win7/8/10 Only) System Hive: (CurrentControlSet\Enum\USBSTOR\Ven_Prod_Version\USBSerial\Properties) (83da6326-97a6-4088-9453-a19231573b29)#### 0064 = First Install (Win7-10) 0066 = Last Connected (Win8-10) 0067 = Last Removal (Win8-10)	Description Discover the Volume Serial Number of the Filesystem Partition on the USB (NOTE: This is not the USB Unique Serial Number, which is hardcoded into the device firmware.) Location - SOFTWARE\Microsoft\Windows NT\CurrentVersion\ENDMgmt - Use Volume Name and USB Unique Serial Number to: - Find last integer number in line - Convert Decimal Serial Number into Hex Serial Number Interpretation - Knowing both the Volume Serial Number and the Volume Name, you can correlate the data across SHORTCUT File (LNK) analysis and the RECENTDOCS key. - The Shortcut File (LNK) contains the Volume Serial Number and Name - RecentDocs Registry Key, in most cases, will contain the volume name when the USB device is opened via Explorer	Description Shortcut files automatically created by Windows - Recent Items - Open local and remote data files and documents will generate a shortcut file (.lnk) Location XP: %USERPROFILE%\Recent Win7/8/10: %USERPROFILE%\AppData\Roaming\Microsoft\Windows\Recent Interpretation - Date/Time file of that name was first opened - Creation Date of Shortcut (LNK) File - Date/Time file of that name was last opened - Last Modification Date of Shortcut (LNK) File - LNKTarget File (Internal LNK File Information) Data: - Modified, Access, and Creation times of the target file - Volume Information (Name, Type, Serial Number) - Network Share information - Original Location - Name of System
User		
Description Find User that used the Unique USB Device. Location - Look for GUID from SYSTEM\MountedDevices - NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 Interpretation This GUID will be used next to identify the user that plugged in the device. The last write time of this key also corresponds to the last time the device was plugged into the machine by that user. The number will be referenced in the user's personal mountpoints key in the NTUSER.DAT Hive.		

Browser Usage

History	Cache	Session Restore
Description Records websites visited by date and time. Details stored for each local user account. Records number of times visited (frequency). Also tracks access of local system files. Location Internet Explorer - IE6-7: %USERPROFILE%\Local Settings\History\History.IE5 - IE8-9: %USERPROFILE%\AppData\Local\Microsoft\Windows\History\History.IE5 - IE10, 11, Edge: %USERPROFILE%\AppData\Local\Microsoft\Windows\WebCache\WebCacheV9.dat Firefox - XP: %USERPROFILE%\Application Data\Mozilla\Firefox\Profiles\<random text>.default\places.sqlite - Win7/8/10: %USERPROFILE%\AppData\Roaming\Mozilla\Firefox\Profiles\<random text>.default\places.sqlite Chrome - XP: %USERPROFILE%\Local Settings\Application Data\Google\Chrome\User Data\Default\History - Win7/8/10: %USERPROFILE%\AppData\Local\Google\Chrome\User Data\Default\History	Description - The cache is where web page components can be stored locally to speed up subsequent visits - Gives the investigator a "snapshot in time" of what a user was looking at online - Identifies websites which were visited - Provides the actual files the user viewed on a given website - Cached files are tied to a specific local user account - Timestamps show when the site was first saved and last viewed Location Internet Explorer - IE8-9: %USERPROFILE%\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5 - IE10: %USERPROFILE%\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5 - IE11: %USERPROFILE%\AppData\Local\Microsoft\Windows\NetCache\IE - Edge: %USERPROFILE%\AppData\Local\Packages\microsoft.microsoftedge_8P59Z4668\MicrosoftEdge\Cache Firefox - XP: %USERPROFILE%\Local Settings\Application Data\Mozilla\Firefox\Profiles\<randomtext>.default\Cache - Win7/8/10: %USERPROFILE%\AppData\Local\Mozilla\Firefox\Profiles\<randomtext>.default\Cache Chrome - XP: %USERPROFILE%\Local Settings\Application Data\Google\Chrome\User Data\Default\Cache - data_# and f_##### - Win7/8/10: %USERPROFILE%\AppData\Local\Google\Chrome\User Data\Default\Cache - data_# and f_##### Flash & Super Cookies Description Local Stored Objects (LSOs), or Flash Cookies, have become ubiquitous on most systems due to the extremely high penetration of Flash applications across the Internet. They tend to be much more persistent because they do not expire, and there is no built-in mechanism within the browser to remove them. In fact, many sites have begun using LSOs for their tracking mechanisms because they rarely get cleared like traditional cookies. Location Win7/8/10: %APPDATA%\Roaming\Macromedia\FlashPlayer\SharedObjects\<randomp offid> Interpretation - Websites visited - User account used to visit the site - When cookie was created and last accessed	Description Automatic Crash Recovery features built into the browser. Location Internet Explorer Win7/8/10: %USERPROFILE%\AppData\Local\Microsoft\Internet Explorer\Recovery Firefox Win7/8/10: %USERPROFILE%\AppData\Roaming\Mozilla\Firefox\Profiles\<randomtext>.default\sessionstore.js Chrome Win7/8/10: %USERPROFILE%\AppData\Local\Google\Chrome\User Data\Default\ Files = Current Session, Current Tabs, Last Session, Last Tabs Interpretation - Historical websites viewed in each tab - Referring websites - Time session ended - Modified time of .dat files in LastActive folder - Time each tab opened (only when crash occurred) - Creation time of .dat files in Active folder Google Analytics Cookies Description Google Analytics (GA) has developed an extremely sophisticated methodology for tracking site visits, user activity, and paid search. Since GA is largely free, it has a commanding share of the market, estimated at over 80% of sites using traffic analysis and over 50% of all sites. _utma – Unique visitors - Domain Hash - Visitor ID - Cookie Creation Time - Time of 2nd most recent visit - Number of visits _utmb – Session tracking - Domain hash - Page views in current session - Outbound link clicks - Time current session started _utmtz – Traffic sources - Domain Hash - Last Update time - Number of visits - Number of different types of visits - Source used to access site - Google Adwords campaign name - Access Method (organic, referral, cpc, email, direct) - Keyword used to find site (non-SSL only)